

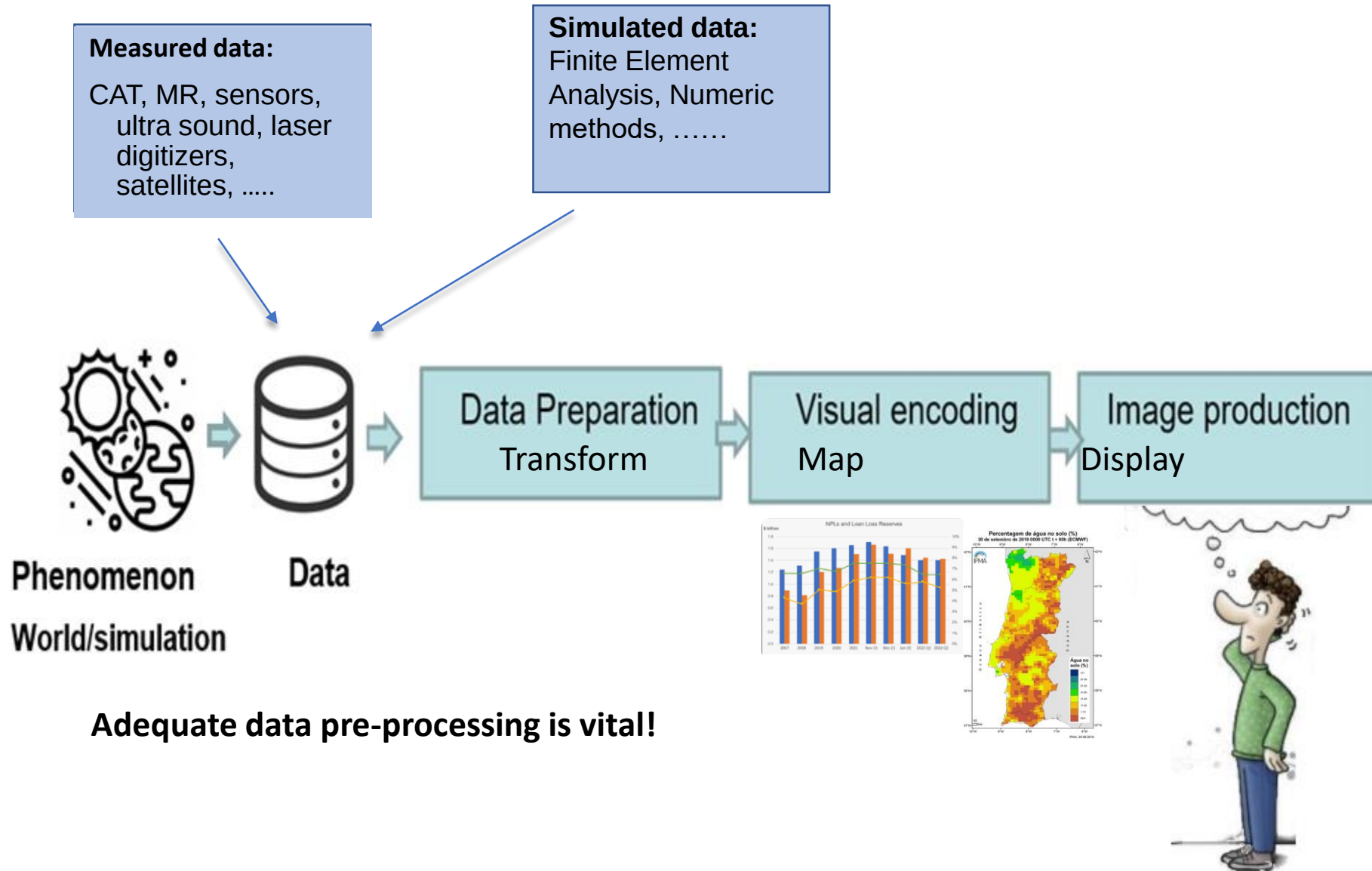


Data



Anomaly Detection

Data is a proxy to the phenomena to analyse and understand



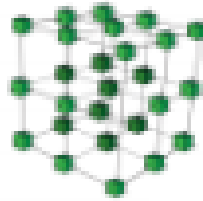
- Data may have a lot of different forms and there are many techniques and systems to visualize them
- A data classification is important to:
 - predict what visualization techniques are adequate
 - make easier the communication about the data
 - allow a more systematic approach to Visualization
 -

name	rank	gender	year
Jacob	1	boy	2010
Isabella	1	girl	2010
Ethan	2	boy	2010
Sophia	2	girl	2010
Michael	3	boy	2010

Data Abstraction

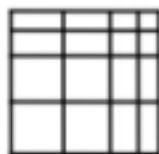
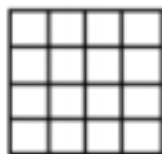
- Four basic dataset types:

- Tables
- Networks
- Fields
- Geometry

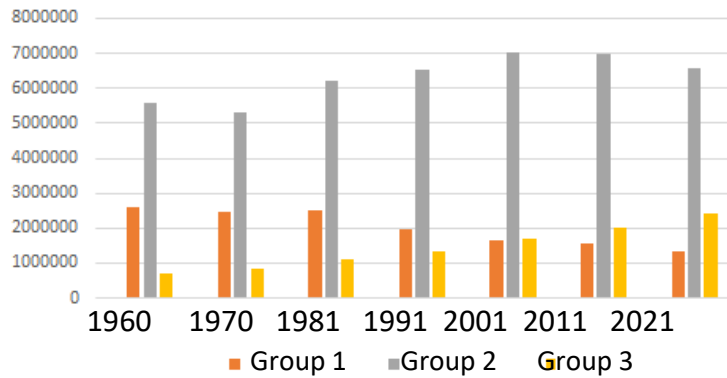


- Five basic datatypes (what may be visualized)

- Items
 - Attributes
 - Links
 - Positions
 - Grids
- Categorical
 Ordered Ordinal
 Quantitative



Population by age group



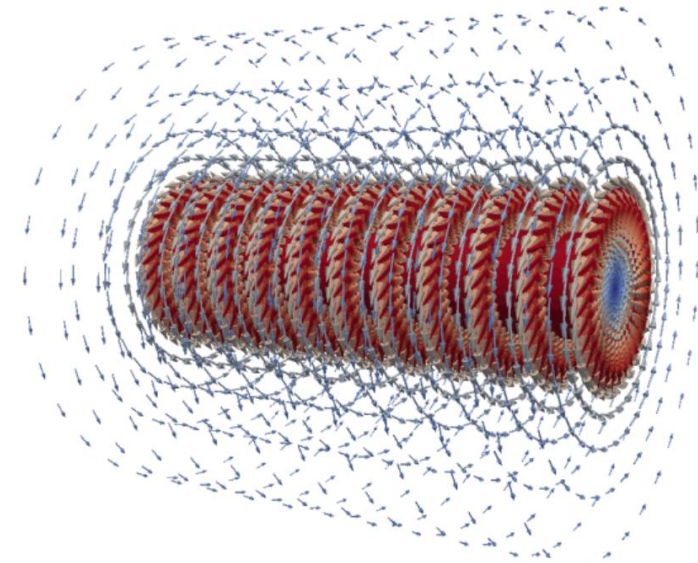
Attribute/variable

Census Year	Population by age group		
	Group 1: 0-14 years	Groups 2: 15-64 years	Group 3: 65 + years
1960	2591955	5588868	708569
1970	2451850	5326515	832760
1981	2508673	cell 6198883	1125458
1991	1972403	6552000	1342744
2001	1656602	7006022	1693493
2011	1572329	6979785	2010064
2021	1331188	6588239	2423639

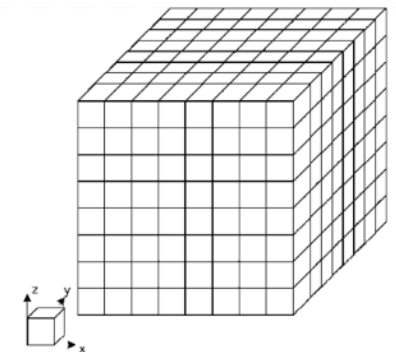
Item/
object

Tabular Data -> InfoVis

3D/4D Spatial Data -> SciVis



[ParaView - Open-source, multi-platform data analysis and visualization application](#)



- **Data representation level:**
 - Qualitative (or categorical)
 - Quantitative (or numeric)

- **Data nature:**
 - Continuous
 - Discrete

Computer data are discrete
but the phenomena may be continuous

- **Measuring scale:**
 - Nominal
 - Ordinal
 - Interval
 - Ratio

>

categorical

>

quantitative

4.1 27 102 3.14
-0.1 16

Numerical data



Categorical data

Monday Wednesday
Tuesday Thursday

Ordinal data

(Spence, 2007)

- What data are obtained from a survey using a Likert-type scale?

“How do you rate this product?”

“How satisfied are you with our service?”

“How do you grade this presentation?”

The image displays three different visual formats for a 5-point Likert-type scale, all set against a light beige background.

- Star Rating Format:** The first panel shows five rows. Each row starts with a small, empty rounded square checkbox, followed by a series of yellow stars. The number of stars decreases from five in the top row to one in the bottom row.
- Descriptive Label Format:** The second panel shows five rows. Each row consists of a descriptive label followed by an empty rounded square checkbox. The labels, from top to bottom, are: "Very satisfied", "Satisfied", "Neutral", "Unsatisfied", and "Very unsatisfied".
- Numeric Rating Format:** The third panel shows five rows. Each row consists of a numeric rating followed by an empty rounded square checkbox. The ratings, from top to bottom, are: "5", "4", "3", "2", and "1".

- Qualitative?
- Ordinal?
- Quantitative (or numeric)?

- Examples of measuring scales and types of data:

- **nominal** --> car brands, gender, animal species...
- **ordinal** --> week days, preferences, levels measured in a Likert-type scale
- **Interval** --> date, IQ, temperatures in °C (have an arbitrary zero)
- **Ratio** --> temperatures in °K, weight, height



- The ratio scale represents the **highest level of representation**, has a non-arbitrary/true zero (unlike the interval scale); ratios are meaningful
- This is a general classification and might be used to select the statistical methods to use with the data

- In what scale are the following variables measured?

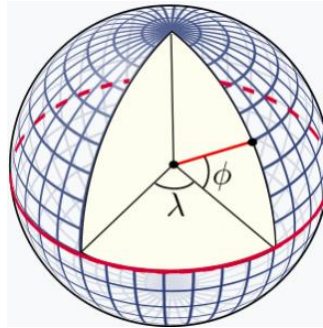
Length (m)



Weight (kg)



Latitude ($^{\circ}$ W/E)



Longitude ($^{\circ}$ S/N)

Height (m)

Temperature ($^{\circ}$ F; $^{\circ}$ C)



Pressure (Pa)



Example: beyond the structure of the data to Visualize: look at the phenomenon

- Consider a data set with three columns:

latitude *longitude* *d*



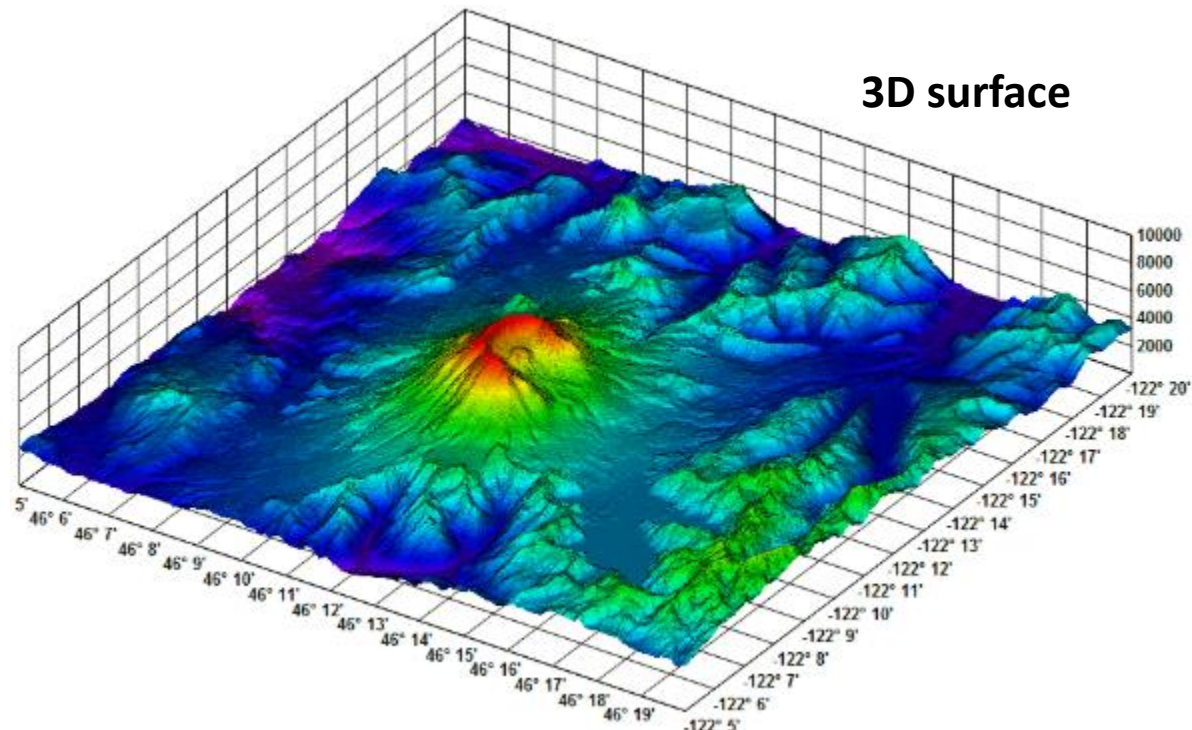
Iso-contours

- Which is the most adequate way to visualize these data?

- If *d* is depth or altitude?

the selected visualization technique may involve interpolation (continuous phenomenon)

(e.g. isocontours, isosurfaces, 3D surface)



3D surface

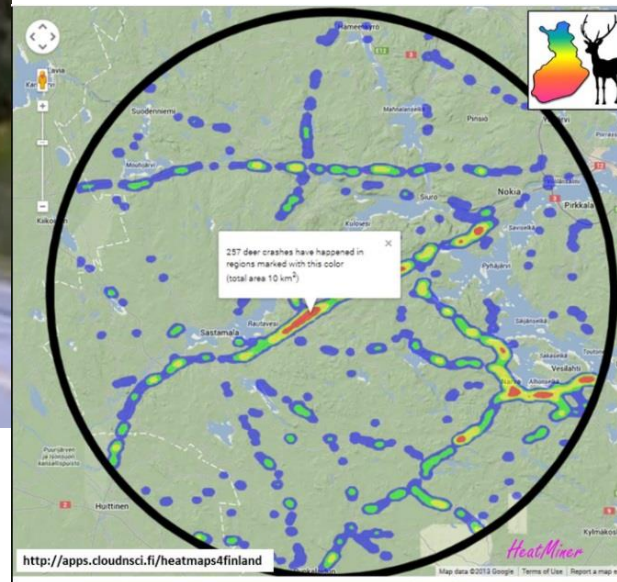
latitude longitude d

- What if data represent location and the number of “deer crash” accidents?



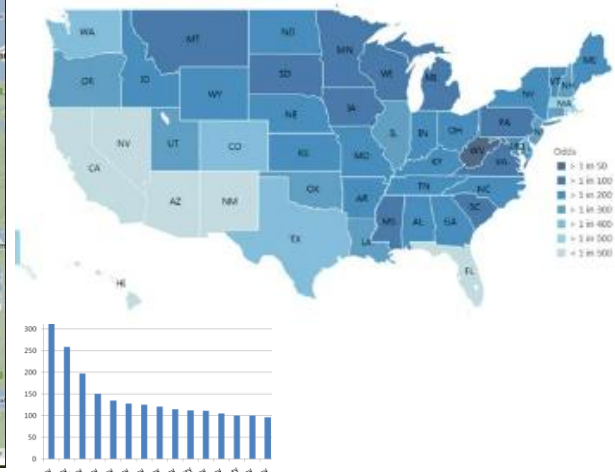
<http://cloudnsi.fi/wiki/index.php?n=Applications.Heatmaps4Finland>

Heat map



Choropleth

Deer vs. Vehicle collisions



Interpolation and contours don't make sense

Know the data structure is not enough

It is necessary to know the phenomenon behind the data as well as knowing the tasks (questions) of the users!



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Data preparation

- Data preparation is **very important and very time consuming**
- Several phases and terms:
 - Data pre-processing
 - Data wrangling
 - Data cleaning, Data tidying ...
 - Data transformation

Data integrity becomes more essential when the volume of data increases

“Brilliant visualizations cannot redeem bad data!”

Or

“Garbage in garbage out ...”

Cleansing Data

- Data is dirty: it contains typos, inconsistencies, outliers, missing data, fails in some way to meet a standard...

Transforming Data

(at the variable level)

- Encoding
- Aggregation
- Derived data
- Removal
- Standardization

Revisiting previous examples:

Max and Min temperatures along the month of February (in °C):

day	Max T	Min. T
1	15	7
2	14	8
3	13	6
4	13	6
5	12	6
6	13	7
7	13	7
8	14	8
9	15	5
10	12	5
11	13	6
12	12	7
13	11	8
14	11	8
15	12	8
16	12	9
17	13	9
18	14	9
19	14	8
20	13	8
21	13	8
22	12	7
23	12	7
24	11	7
25	11	6
26	11	7
27	13	6
28	14	6

Q4- How were the daily temperature amplitudes?

Q5 – What was the maximum temperature amplitude?

- Should we use a derived variable to answer Q4 and Q5?
- What if we are addressing an audience in the USA?
Should we use some other temperature unit?

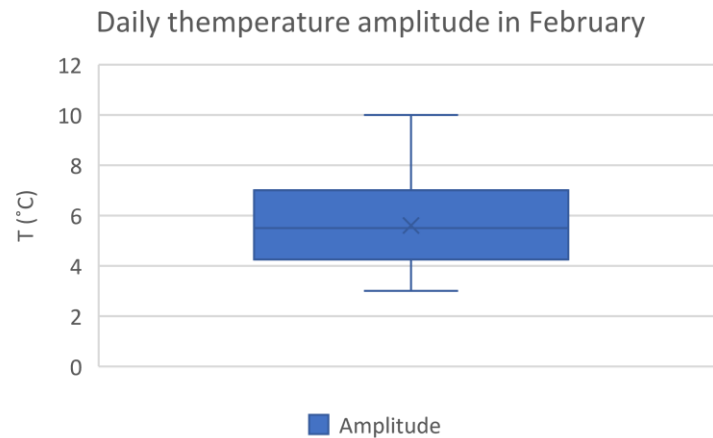
Revisiting a previous example:

Daily temperature amplitudes along the month of February (in °C):

Day	Amplitude
1	8
2	6
3	7
4	7
5	6
6	6
7	6
8	6
9	10
10	7
11	7
12	5
13	3
14	3
15	4
16	3
17	4
18	5
19	6
20	5
21	5
22	5
23	5
24	4
25	5
26	4
27	7
28	8

Q4- How were the daily temperature amplitudes?

Q5 – What was the maximum temperature amplitude?



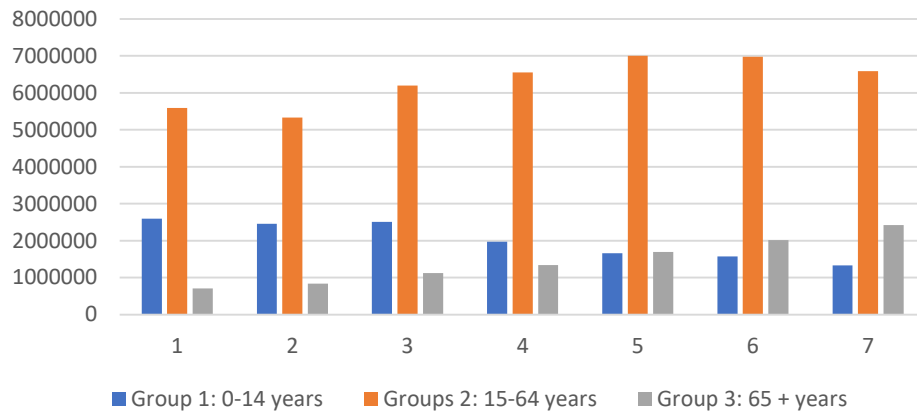
Another example:

Census Year	Population by age group		
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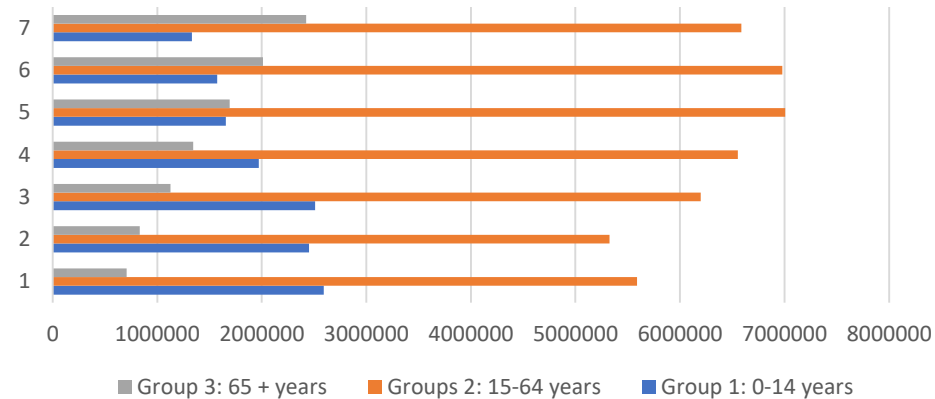
- What if we are studying the evolution of the senior population?
- Or only children?
- Or everyone?

studying the entire population:

Age groups - Census from 1960-2021
(inhabitants)

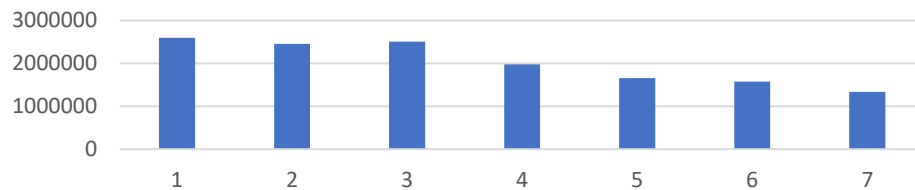


Age groups - Census from 1960-2021
(inhabitants)



studying only children:

Group 1: 0-14 years (1960-2021)



More examples:

Cleaning Data

Birth date: Feb/30/2000

Temperature: -300 °K

City: Lixboa

Transforming Data

- Encoding – answers to an open question need to be parsed and coded
- Aggregation – detail may be excessive (age: <18; 18-40; 41-65; >65)
- Derived data – add new relevant variables ($T_{range} = T_{max} - T_{min}$)
- Removal – remove data that are not needed
- Standardization – M/F; °C or °F

Main bibliography

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